PHASE REPORT / PERSONAL TECHNICAL ARTIFACT

TRISK-POWER-PILOT

A compact artifact of work completed, decisions made, and follow-on paths worth exploring.

project :: PACTA Vietnam

date :: 2026-04-16

repo :: pacta-vietnam

PROJECT

PACTA Vietnam

REPOSITORY

pacta-vietnam

DATE

2026-04-16

01 / INPUT -> OUTPUT

Input -> Output Summary

INPUTS

- Stabilized Vietnam power outputs from the completed PACTA baseline
- Synthetic assumptions documented in `docs/TRISK_Demo_Assumptions.md`
- Installed `trisk.model` package with folder inputs generated by `scripts/trisk_prepare_inputs.R`

OUTPUTS

- Runnable TRISK input package in `output/trisk_inputs/power_demo/`
- Package-backed stress results in `synthesis_output/trisk/power_demo/`

- Borrower prioritization table combining NPV, PD, and PACTA alignment gap in `top_borrowers_alignment_trisk.csv`

02 / LOGIC FLOW

Logic Flow

03 / MATH / ALGORITHM USED

Math / Algorithm Used

- The bridge back-calculates `capacity` so that `capacity * capacity_factor` reproduces the repo's existing synthetic production plan for each power technology-year row.
- The package-backed TRISK run applies discounted cash flow logic for NPV change and a Merton-style credit-risk step for PD change.
- Borrower prioritization adds a lightweight composite score: 70% scaled NPV deterioration, 20% scaled PD change, and 10% average absolute PACTA alignment gap.

04 / TOOLS / FORMULAS / METHODS

Tools / Formulas / Methods

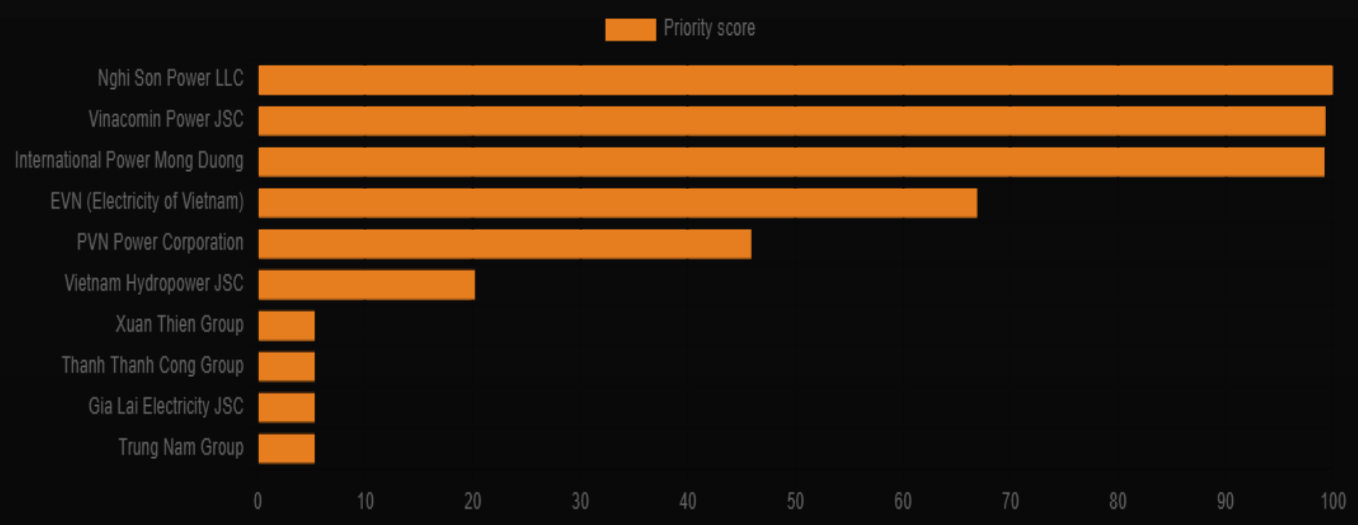
ITEM	DETAIL
Assumptions note	docs/TRISK_Demo_Assumptions.md

ITEM	DETAIL
Input builder	scripts/trisk_prepare_inputs.R
Pilot runner	scripts/trisk_power_demo.R
Package	trisk.model 2.6.1
Run parameters	{"baseline_scenario": "VN_PDP8_BASELINE", "target_scenario": "VN_NZE_STRESS", "scenario_geography": "Vietnam", "carbon_price_model": "increasing_carbon_tax_50", "risk_free_rate": "0.03", "discount_rate": "0.08", "growth_rate": "0.02", "div_netprofit_prop_coef": "1", "shock_year": "2028", "market_passthrough": "0.25", "run_id": "d744d883-b8a6-409d-ab65-b79434c05b41"}

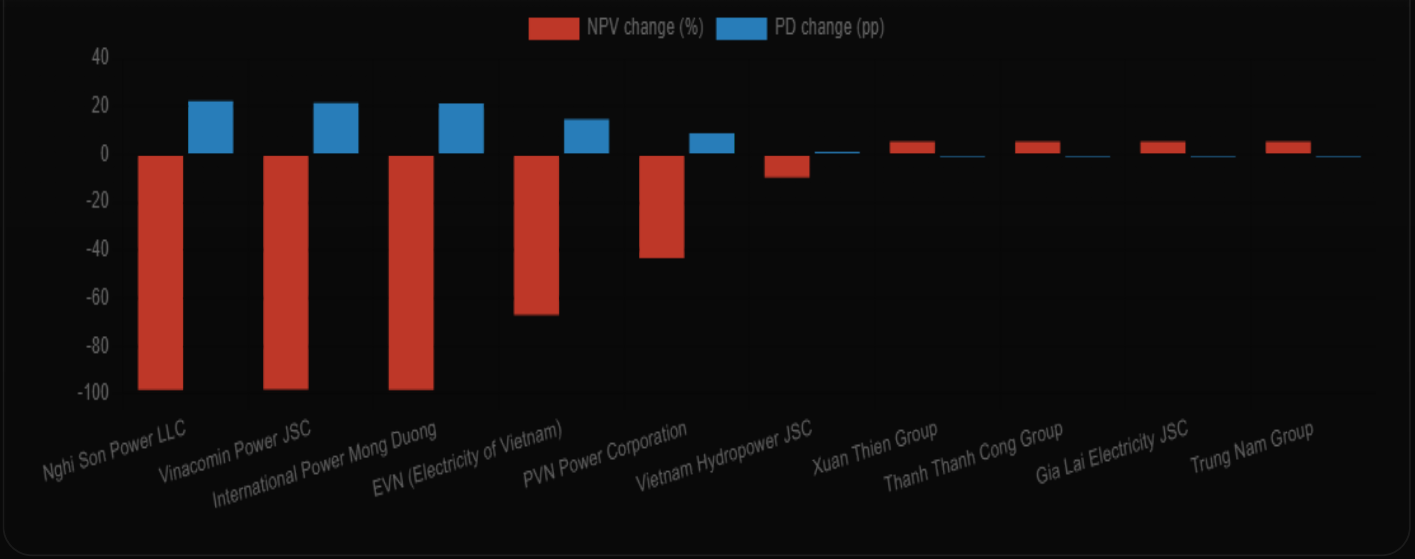
05 / RESULTS & VISUALIZATIONS

Results & Visualizations

Top Borrower Stress Priority Score



NPV vs PD Stress Change for Top Borrowers



06 / LIMITATIONS / SECOND-BEST ALTERNATIVE

Limitations / Second-Best Alternative

- The pilot only covers the power sector, which is deliberate for first-pass credibility but not yet a whole-portfolio TRISK view.
- Financial features, price paths, and carbon prices are synthetic demo inputs, even though the package execution is real.
- Second-best alternative would have been a custom pseudo-TRISK scoring model without the package, but the successful package-backed run is materially stronger evidence.

07 / ERRORS / WARNINGS / FLAGS

Errors / Warnings / Flags Encountered

- The first bridge attempt duplicated scenario rows by company, which produced list-columns inside trisk.model; fixing the scenario file to technology-year granularity resolved this.
- The PD stage removed 5 rows during Merton compatibility checks, but the run still completed and returned borrower-level results.
- Dung Quat LNG Power Consortium currently lands at zero baseline output in the summary and should be treated as an edge case for future refinement.

08 / OPEN QUESTIONS / NEXT PHASE SEEDS

Open Questions / Next Phase Seeds

- Should the next iteration add systematic sensitivity runs for `shock_year`, `discount_rate`, and `market_passthrough`?
- Should gas and renewable price paths be calibrated to a more explicit Vietnam market story before sharing the pilot broadly?
- Is the next highest-value extension a multi-sector TRISK expansion or a tighter power-sector calibration with better financial proxies?